Data Analysis SAC Part A Practise Suggested Solutions

Immunisation is a key aspect of public health policy for disease control around the world. In particular, data on immunisation rates is an important part of analysis. In this task we investigate the percentage of children aged 12-23 months immunised against the disease of measles.

How have the immunisation percentages changed for children between 1990 and 2014?

For this investigation use the data from Table 1.

a. Complete a univariate analysis of the immunisation percentages for children for 1990 and 2014. Include in your analysis any graphs, diagrams or box plots that you decide would be of benefit to you in highlighting differences and similarities between the two years. Discuss why you decided to use your chosen graph(s) and displays for the data set.

b. Compare the two distributions in terms of shape, centre, spread and outliers. Discuss whether an association exists between the immunisation percentages for children over the two years.

Table 1: The percentage of children aged 12-23 months who received the measles vaccination in 2014 compared to 1990 in 30 countries.

Country Name	Hemisphere	% of children immunised 1990	% of children immunised 2014
Afghanistan	Northern	20	66
Argentina	Southern	93	95
Australia	Southern	86	93
Brazil	Southern	78	99
Cambodia	Northern	34	80
Canada	Northern	89	90
Colombia	Northern	82	91
Costa Rica	Northern	90	95
Denmark	Northern	84	90
Ethiopia	Northern	38	70
Fiji	Southern	84	94
Greece	Northern	76	97
Grenada	Northern	85	94
Guinea	Northern	35	52
Hungary	Northern	99	99
India	Northern	56	85
Ireland	Northern	78	93
Mali	Northern	43	80
Mauritius	Southern	76	98
Mexico	Northern	75	97
Morocco	Northern	79	99
Nepal	Northern	57	88
New Zealand	Southern	90	93
Samoa	Southern	89	82
Singapore	Northern	84	95
Switzerland	Northern	90	93
Thailand	Northern	80	99
United Kingdom	Northern	87	93
United States	Northern	90	92
Vietnam	Northern	88	97

Your investigation may include the following:

1. Complete a univariate analysis of the immunisation percentages for children for 1990 and 2014.

a. Classify the data using categorical nominal, categorical ordinal, numerical discrete and numerical continuous.

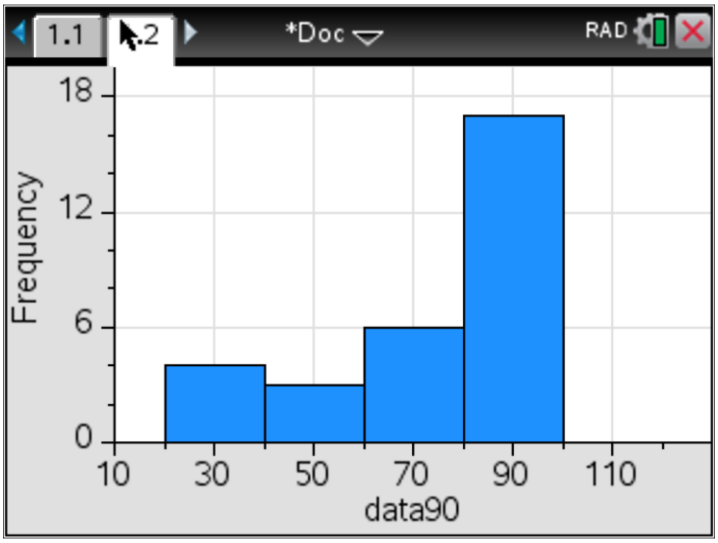
Country name: Categorical nominal

Percentage of children immunised: *Numerical discrete*

b. Complete the frequency table, summarising the data collected for the *percentage of children in 1990*.

% of children	**Frequency**
0 –	0
20 –	4
40 -	3
60 –	6
80 –	17
TOTAL:	30

c. Using the frequency table from part **b**, prepare a graphical display showing the percentage of children immunised in 1990. (Hint: draw a histogram to display the percentage of children immunised in 1990.)



d. Use the histogram to describe the distribution, centre and spread of the data.

The histogram shows that the distribution for the percentage of children immunised in 1990 is negatively skewed with no outliers. The distribution is centred between 80-100%. The distribution has a spread of 80 (100-20) as measured by the range.

e. Would the mean or median be a better measure of central tendency for this data? Explain your answer. What is the value of the appropriate measure of centre (median or mean) for your 1990 data?

The median would be a better measure of the central tendency for the data as the data is skewed. The median value for the percentage of children immunised in 1990 is 83%

f. Is it appropriate to calculate z-scores for the percentage of children immunised in 1990 data? Explain your answer.

No as z scores are only calculated for symmetric data and the data we have is negatively skewed.

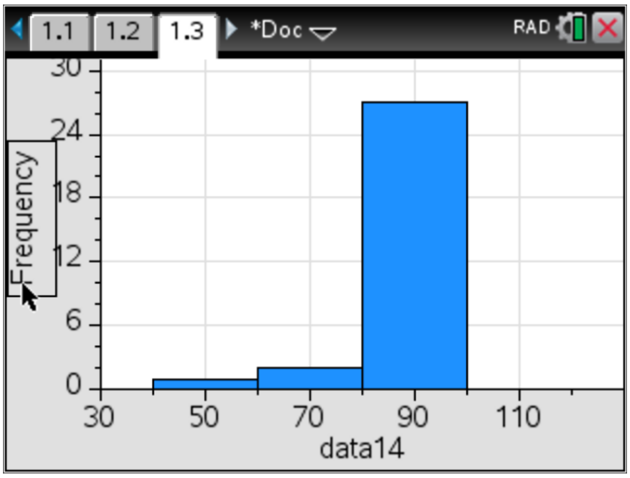
2. Complete a univariate analysis of the immunisation percentages for children for 1990 and 2014.

Repeat **1 b-f** for 2014

b. Complete the frequency table, summarising the data collected for the *percentage of children in 1990*.

% of children	**Frequency**
0 –	0
20 –	0
40 -	1
60 –	2
80 –	27
TOTAL:	30

c. Using the frequency table from part **b**, prepare a graphical display showing the percentage of children immunised in 1990. (Hint: draw a histogram to display the percentage of children immunised in 1990.)



d. Use the histogram to describe the distribution, centre and spread of the data.

The histogram shows that the distribution for the percentage of children immunised in 2014 is negatively skewed with no outliers. The distribution is centred between 80-100%. The distribution has a spread of 60 (100-40) as measured by the range.

e. Would the mean or median be a better measure of central tendency for this data? Explain your answer. What is the value of the appropriate measure of centre (median or mean) for your 2014 data?

The median would be a better measure of the central tendency for the data as the data is skewed. The median value for the percentage of children immunised in 2014 is 93%

f. Is it appropriate to calculate z-scores for the percentage of children immunised in 2014 data? Explain your answer.

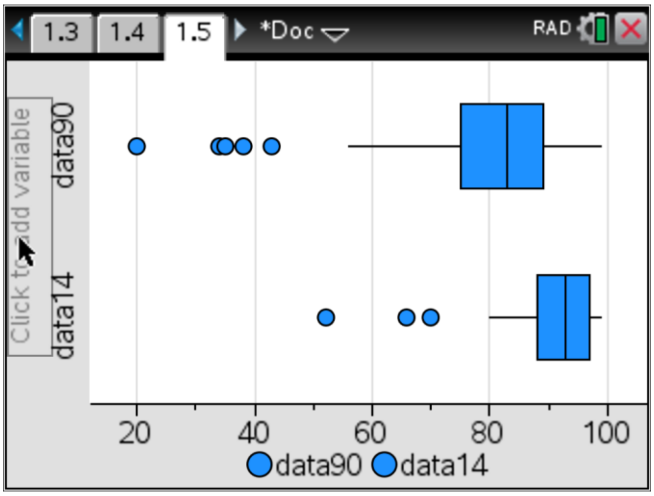
No as z scores are only calculated for symmetric data and the data we have is negatively skewed.

3.
a. Calculate the five number summary for the percentage of children immunised in 1990 and in 2014.

	% of children immunised in 1990	**% of children immunised in 2014**
Min:	20	52
Q1:	75	88
Median:	83	93
Q3:	89	97
Max:	99	99

b. Prepare a graphical display showing the percentage of children immunised in 1990 and in 2014. Discuss why you decided to use your chosen graphical display for the data set.

Parallel boxplots were chosen to display the data there is one categorical variable (*year*) and one numerical variable (*percentage of children immunised*)



(HINT: Using your CAS calculator prepare a display of parallel box plots showing the percentage of children immunised in 1990 and in 2014.)

c. Compare the two distributions in terms of shape, centre, spread and outliers.

The boxplots show that both distributions for the percentage of children immunised in 1990 and 2014 are negatively skewed and both have outliers at the lower end. 2014 data has 3 outliers (52, 66, 70) whereas the 1990 data has 5 outliers (20, 34, 35, 38, 43).

The 2014 distribution has a higher percentage of children immunised with the median value being 93% compared to the 1990 distribution which has a median value of 83%.

The spread of the percentage of children immunised decreases from 1990 to 2014. The IQR in 1990 is 14 (89-75) is larger than in 2014 is 9 (97-88).

d. Does there appear to be an association between the percentage of children immunised in 1990 and in 2014? Discuss whether an association exists between the immunisation percentages for children over the two years giving reference to an appropriate statistic.

Yes there is an association between the percentage of children immunised and years due to the change in the median from 1990 to 2014.

The median was 81% in 1990 which increased to 93% in 2014.

4.

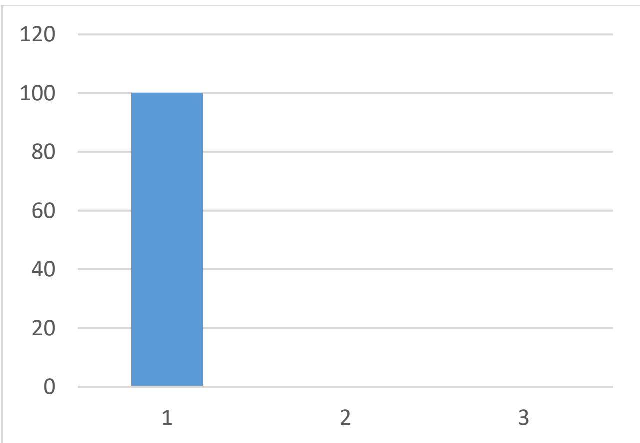
a. Using the 30 countries complete the following two way table using one year of data.
Data used is 1990

Percent immunised	**Hemisphere**	
	Northern	**Southern**
Low (0%-)	2	0
Medium (35%-%)	5	0
High (70%-100%)	16	7
Total	23	7

b. Use the data in the table above to complete the percentage two-way table.

Percent immunised	**Hemisphere**	
	Northern	**Southern**
Low (0%-)	8.7	0
Medium (35%-)	21.7	0
High (70%-100%)	69.6	100
Total	100	100

c. Prepare a graphical display for this two way frequency table. Discuss why you decided to use your chosen graphical display for the data set. (HINT: Convert this data into a segmented bar chart)



d. Is there an association between the percentage of children immunised and the hemisphere in which they live? Write a brief response quoting appropriate statistics.

For the 1990 data there is an association between the percentage of children immunised and the hemisphere in which they live. In the Northern hemisphere 69.6% of the countries had a LOW percentage immunised whereas in the Southern hemisphere 100% of the countries had a HIGH percentage immunised rate.

The percentage of countries in the Northern hemisphere with a LOW immunisation percentage is less than the countries in the Southern hemisphere.

5.
Using the statistical information you have gathered by completing this investigation, discuss the immunisation rates in the 30 countries in 1990 as compared to 2014.

Students to summarise the information from their investigation in a short paragraph.